

Session 4.1: Hierarchical Models, Priors and Model Checking

Imperial College London

Learning Objectives

After this session you should be able to:

- Understand the different modelling assumptions for hierarchical data
- Be able to specify a hierarchical model for Poisson data
- Be able to perform prediction in a Bayesian approach
- Distinguish and choose between several prior distributions for the precision/variance parameter
- Use the WAIC as tool for model selection.

The topics treated in this lecture are covered in Chapter 5 of Blangiardo and Cameletti (2015).

Outline

1. What are hierarchical models
2. Different modelling assumptions
3. Parameter interpretation
4. Hierarchical regression
6. Choice of prior
7. Model selection

What are hierarchical models

What are hierarchical models?

Hierarchical model is a very broad term that refers to wide range of model set-ups

- Multilevel models
- Random effects models
- Random coefficient models
- Variance-component models
- Mixed effect models

Key feature: Hierarchical models are statistical models that provide a formal framework for analysis with a complexity of structure that matches the system being studied.

The hierarchical approach

- Attempt to capture (model) and understand the structure of the data
- Is flexible:
 - all sources of correlation and heterogeneity can be incorporated in a modular fashion, in particular by the introduction of unit-specific parameters
 - can be combined with other types of models, e.g. for missing data or measurement error
- We wish to make inference on models with many parameters $(\lambda_1, \dots, \lambda_N)$ measured on N units (individuals, areas, time-points, etc.) which are related or connected by the structure of the problem.
- Unit specific parameters will **borrow strength** from corresponding parameters associated with the other units

Motivating example: Disease mapping

- To summarise spatial and spatio-temporal variation in disease risk
- **Question:** Which areas have particularly high or low disease rates?
- **Question:** Can we explain some of the variation in disease rates by area-level covariates?
- Data are the observed (y_i) and expected number of cases in area i : $E_i = \sum_k n_{ik} r_k$, where r_k reference rate for stratum k (age, sex,...)
- Rare disease and/or small areas: Poisson framework

$$y_i \sim \text{Poisson}(\rho_i E_i)$$

where ρ_i is the **unknown RR** in area i

Non smoothed estimates of the RR (SMR or SIR)

$$\text{SMR}_i = \frac{y_i}{E_i}$$
$$\text{Var}(\text{SMR}_i) = \frac{y_i}{E_i^2}$$

- **very imprecise: areas with small E_i have high associated variance**
- **estimated independently: makes no use of risk estimates in other areas of the map**

Motivating example: Disease mapping

Example:

- observed cases of lip cancer y_i diagnosed in Scotland in 1975-1980 at county level $i = 1, \dots, 56$ areas
- expected number of cases E_i are also available using age/sex standardised reference rates and population counts:

Assume a Poisson likelihood for the disease counts in each area:

$$y_i \sim \text{Poisson}(\lambda_i) \qquad \lambda_i = \rho_i E_i \qquad i = 1, \dots, 56$$

- We have 56 parameters ρ_i (one for each area). What prior do we specify on ρ_i ?

Expected numbers of cases - definition

- Expected number of cases if the population had the same stratum-specific mortality/incidence rates as in a reference area
- Adjustments (strata): age, gender ...

Indirect standardisation: $E_i = \sum_k n_{ik} r_k$ with

- r_k : disease rate for stratum k in the reference population
- n_{ik} : population at risk in area i , stratum k

If internal comparison: $\sum_{i=1}^N O_i = \sum_{i=1}^N E_i$

Direct standardisation: apply the disease rate in the population of interest (e.g. UK) to a standard population e.g. European standard population

External comparison: if the reference population is not the population of the study of interest. For example, to calculate the expected numbers in London, risks in England could be used.

Expected numbers of cases - calculation

Strata	Reference area=EW			Ward A		
Age group	Population	Observed	Age-specific rate per 100,000 males	Population	Observed	Expected
	n_k	O_k	$r_k = \frac{O_k}{n_k}$	n_{ik}	O_{ik}	$E_{ik} = \frac{n_{ik} * r_k}{100000}$
0–4	41,400,692	15	0.04	11,438	0	0.00
5–9	41,143,722	6	0.01	9,697	0	0.00
10–14	41,469,696	9	0.02	9,026	0	0.00
15–19	43,087,823	39	0.09	8,650	0	0.01
20–24	45,441,353	79	0.17	12,409	0	0.02
25–29	46,873,725	172	0.37	16,963	0	0.06
30–34	46,927,658	518	1.10	17,303	0	0.19
35–39	46,936,367	1,465	3.12	13,847	0	0.43
40–44	45,304,711	4,136	9.13	11,843	1	1.08
45–49	41,657,557	9,835	23.61	9,457	5	2.23
50–54	38,451,416	20,929	54.43	8,561	3	4.66
55–59	35,842,426	40,427	112.79	7,613	8	8.59
60–64	32,480,032	68,230	210.07	6,968	5	14.64
65–69	28,231,499	95,794	339.32	6,290	15	21.34
70–74	23,315,240	110,371	473.39	5,098	27	24.13
75–79	17,297,264	102,038	589.91	4,049	22	23.89
80–84	10,498,214	68,273	650.33	2,616	20	17.01
85+	6,289,452	38,748	616.08	1,312	12	8.08
TOTAL	632,648,846	561,084		163,140	118	126.38

$$SIR_A = \frac{118}{126.38} = 0.93$$

- Fewer incident cases of lung cancer for males in ward A than expected in EW after adjusting for differences in age.
- In R we can perform indirect standardization using the package SpatialEpi (we will see it in the Practical in week 6).

Modelling assumptions

Different modelling assumptions

Identical parameters

- Assume $\rho_i = \rho$

↪ all the data can be pooled and the individual areas ignored.

- Assume a prior $\rho \sim \text{Gamma}(1, 1)$

↪ conjugate prior

- One parameter generates all the observations
- Very easy to implement as it is conjugate (no need for INLA) and all the data are **pooled** to produce one estimate of the parameter of interest
- Can be unrealistic (it does not take into account differences in the areas)

Different modelling assumptions

Independent parameters

- All the ρ_i are unrelated, meaning that the areas are analysed independently
- Assume a prior $\rho_i \sim \text{Gamma}(1, 1)$; $i = 1, \dots, 56$

⇒ individual estimates of ρ_i are likely to be highly variable (unless very large sample sizes)

- Every area is treated separately (No exchange of information between these). Estimates close to SMR ($\rho_i \approx y_i / E_i$).
- Again no need for INLA, conjugacy can be exploited.

Different modelling assumptions

Similar (exchangeable) parameters

- All the ρ_i are assumed to be *similar*

↪ they come from the same distribution (are generated by the same parameters)

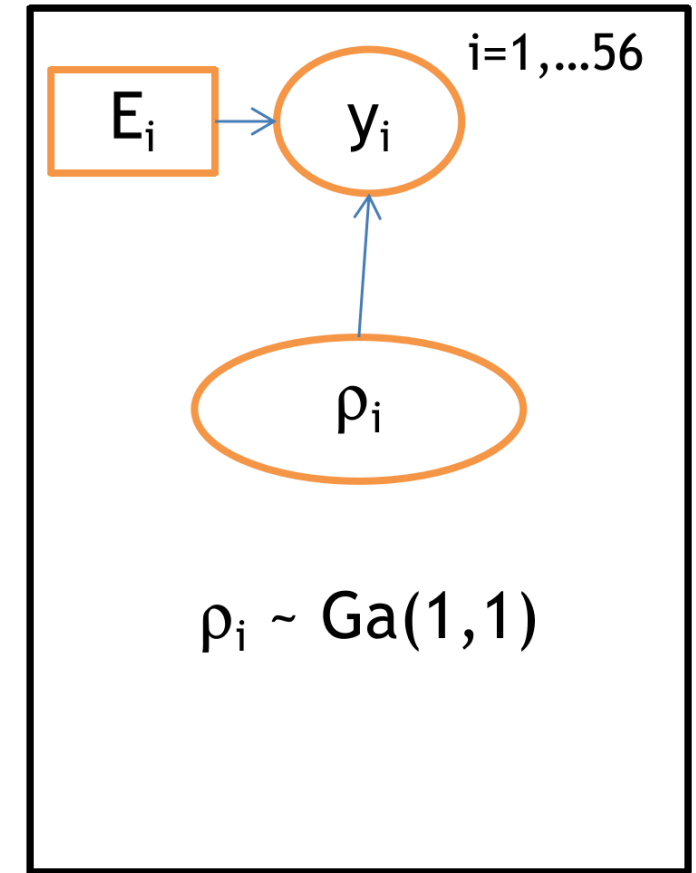
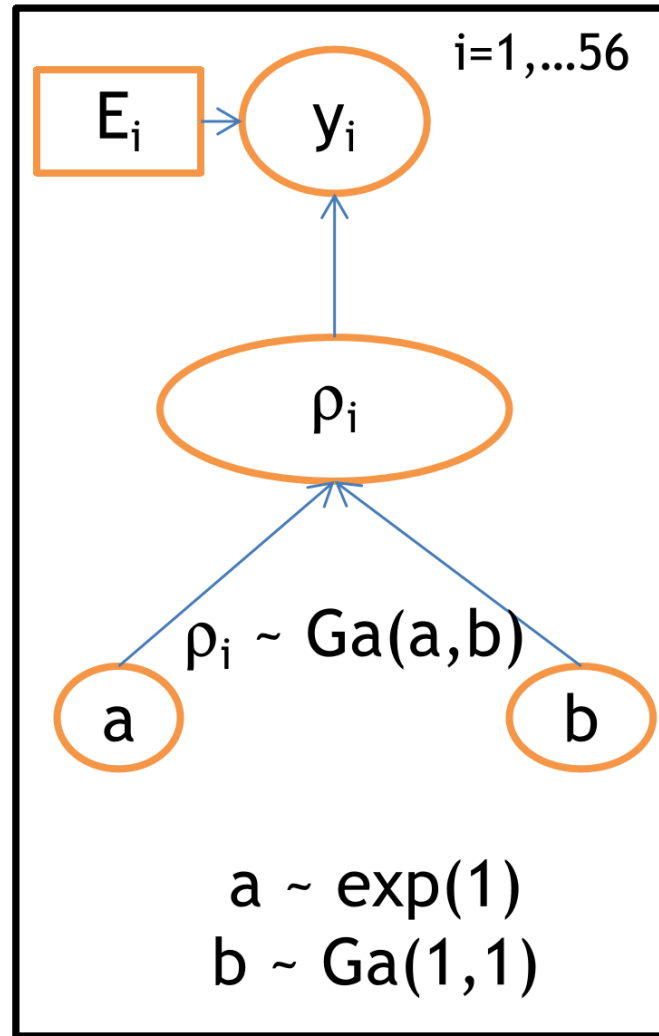
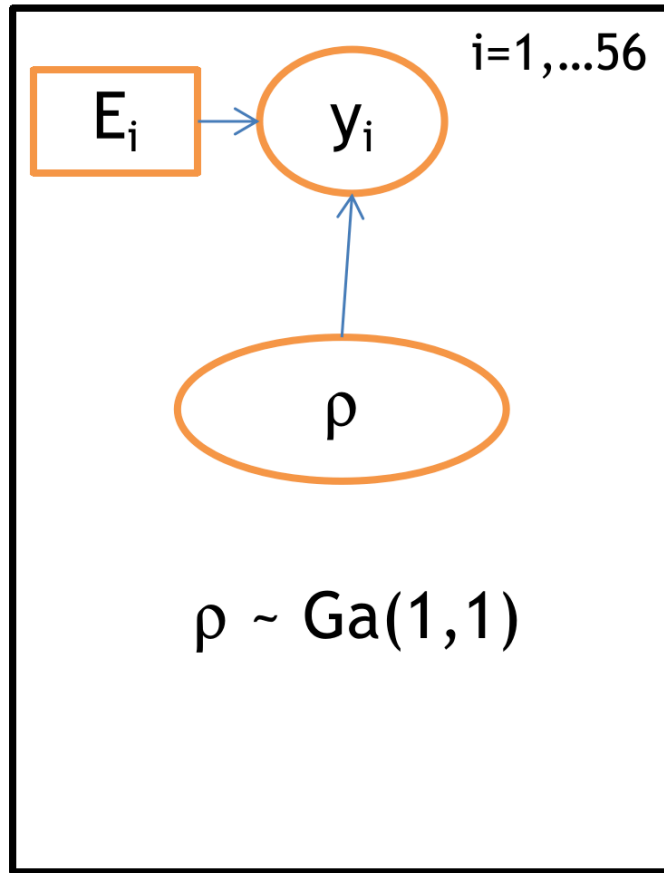
- Assume a hierarchical prior $\rho_i \sim \text{Gamma}(a, b)$

where a and b are unknown parameters and need to be estimated.

- Different levels of analysis
- Allow the exchange of information between different levels as they are all connected to each other
- Assign hyperprior distribution to a and b , for instance

$$a \sim \text{Exp}(1); b \sim \text{Exp}(1)$$

Graphical representation of lip cancer hierarchical model



A more flexible hierarchical prior for the relative risks

- A gamma random effect prior for the ρ_i is mathematically convenient, but might be restrictive:
 - Covariate adjustment is difficult
 - Not possible to allow for spatial correlation between risks in nearby areas
 - A Normal random effect prior on the $\log \rho_i$ is more flexible:

$$y_i \sim \text{Poisson}(\lambda_i = \rho_i E_i)$$

$$\eta_i = \log \rho_i = b_0 + v_i$$

$$v_i \sim \text{Normal}(0, \sigma_v^2)$$

- Need to specify hyperprior distributions for:
 - σ_v^2 (between-area variance), e.g. $1/\sigma_v^2 \sim \text{Gamma}(1, 0.001)$
 - b_0 (mean log relative risk), e.g. $b_0 \sim \text{Normal}(0, 0.0001)$

Advantages of this approach:

Posterior for each v_i

- *borrow strength* from the likelihood contributions of **all** the areas, via their joint influence on the estimate of the unknown population (prior) parameter σ_v^2

→ *global smoothing* of the area RR

→ reflects our *full uncertainty* about the true values of σ_v^2

Interpretation

Parameter interpretation and useful quantities

- ρ_i is the relative risk for the area i compared to the average area with the same structure in the expected values.
- v_i are the random effects. It can also be seen as the latent variable which captures the effect of unknown or unmeasured area level covariates.
- If area level covariates are spatially structured we should take this into account when modelling v_i (we will see it later)
- $\exp(v_i)$ relative risk in area i compared to the risk for the whole study region
- The variance of the random effects σ_v^2 reflects the amount of extra-Poisson variation in the data

Lip cancer dataset

```
> LipCancer <- read.csv("scotlip.csv")  
> LipCancer
```

```
# A tibble: 6 × 11
```

	CODENO	AREA	PERIMETER	RECORD_ID	DISTRICT	NAME	CODE	y	POP	E	x
	<int>	<dbl>	<dbl>	<int>	<int>	<chr>	<chr>	<int>	<int>	<dbl>	<int>
1	6126	974002000	184951	1	1	Skye-Lochalsh	w6126	9	28324	1.38	16
2	6016	1461990000	178224	2	2	Banff-Buchan	w6016	39	231337	8.66	16
3	6121	1753090000	179177	3	3	Caithness	w6121	11	83190	3.04	10
4	5601	898599000	128777	4	4	Berwickshire	w5601	9	51710	2.53	24
5	6125	5109870000	580792	5	5	Ross-Cromarty	w6125	15	129271	4.26	10
6	6554	422639000	118433	6	6	Okney	w6554	8	53199	2.4	24

- DISTRICT identifies the area
- y identifies the counts of cancer cases
- E identifies the expected cases of cancer using the entire region under study as reference
- x identifies the exposure to sun (percentage of agriculture , farming and fishery works)

We first populate the formula environment

```
> formula.inla <- y ~ 1 +  
+   f(RECORD_ID,model="iid", hyper=list(prec=list(prior="loggamma",  
+   param=c(1,0.01))))
```

- The model specification is exactly the same as in GLM;
- Anything with `f(.)` specifies a random effect; in this case `iid` represents the exchangeable structure.

Then we run the model through

```
> lipcancer.poisson <- inla(formula.inla,family="poisson",  
+   data=LipCancer, E=E,  
+   control.predictor=list(compute=TRUE),  
+   control.compute=list(config=TRUE),  
+   control.fixed=list(mean.intercept=0,prec.intercept=0.00001))
```

Note that

- `control.fixed` allows to specify the parameters of the prior for the fixed effects (intercept)
- `control.predictor` tells INLA to include the linear predictor estimation (the parameters of the prior for the fixed effects (intercept)) useful for prediction - see later)
- `control.compute` allows to include model selection indexes, as well as to draw samples from the joint posterior

Results for lip cancer in Scotland example

- $\exp(b_0 + v_i)$ is the relative risk of lip cancer in area i relative to the average area with the same age/sex structure (see map)
- σ_v is the between-area standard deviation of log relative risk of lip cancer
- As in INLA we get the precision we need to convert it into standard deviation using

```
> sigma.v<- inla.tmarginal(function(x) sqrt(1/x),  
+                        lipcancer.poisson$marginals.hyperpar[[1]])
```

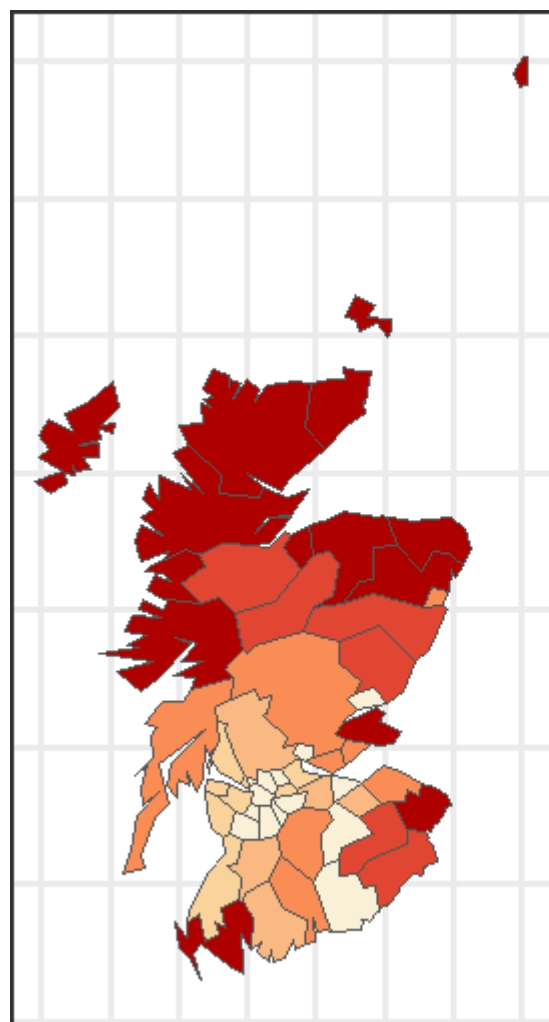
And we can calculate quintiles with

```
> inla.qmarginal(seq(0,1,0.2),sigma.v)
```

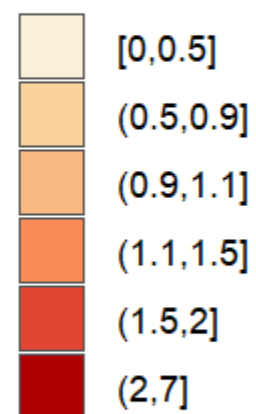
```
[1] 0.5014723 0.6745473 0.7256807 0.7734031 0.8335807 1.1506232
```

Maps: comparing SMR with smoothed estimates

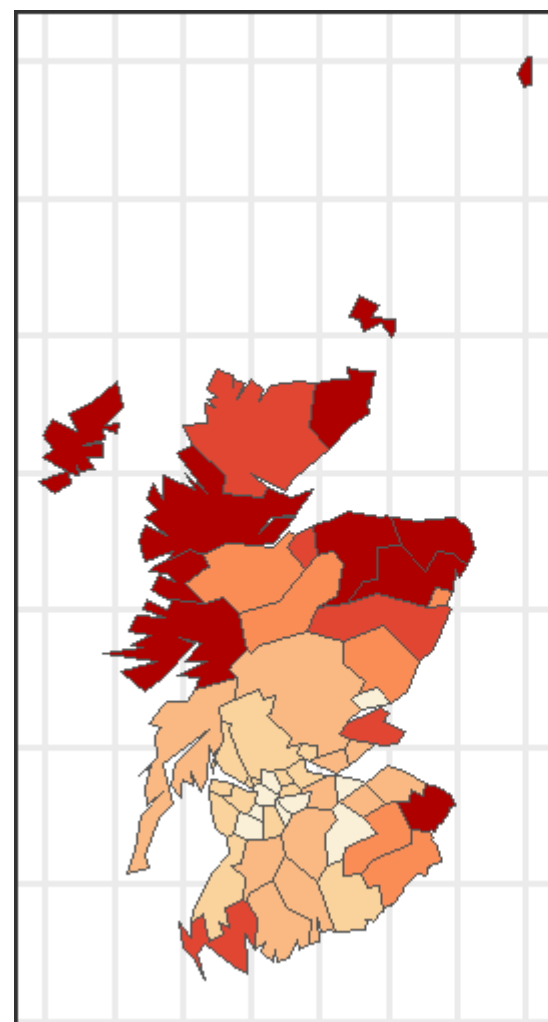
SMR



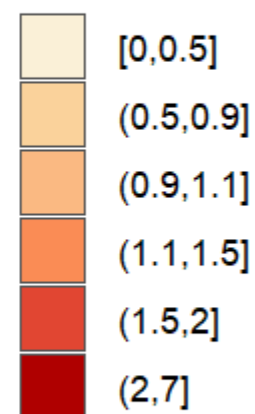
SMR



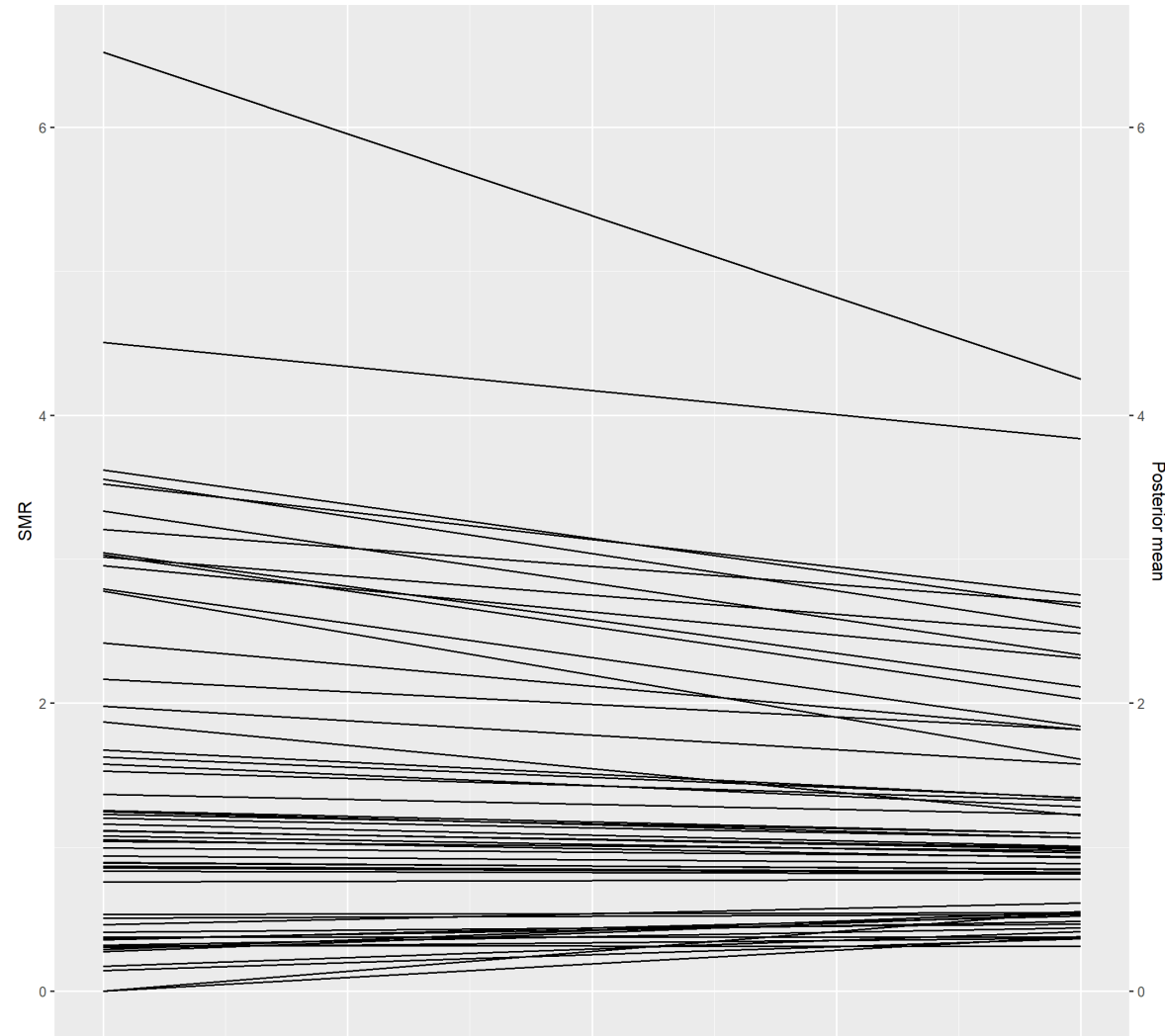
Posterior mean



Post mean



SMR versus posterior mean RR for selected areas



- Comparing the SMR and the area level posterior mean from the model shows a shrinkage towards the global (national mean)

Hierarchical Regression

Regression in INLA

It is easy to move from hierarchical models to regression models with random effects.

Example: Still considering the Lip cancer dataset, we want to evaluate if the variable x , which represents the percentage of agriculture, farming and fishery works, is associated with the outcome. The hypothesis is that exposure to sun could be a risk factor for lip cancer.

A tibble: 6 × 11

	CODENO	AREA	PERIMETER	RECORD_ID	DISTRICT	NAME	CODE	y	POP	E	x
	<int>	<dbl>	<dbl>	<int>	<int>	<chr>	<chr>	<int>	<int>	<dbl>	<int>
1	6126	974002000	184951	1	1	Skye-Lochalsh	w6126	9	28324	1.38	16
2	6016	1461990000	178224	2	2	Banff-Buchan	w6016	39	231337	8.66	16
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6	6554	422639000	118433	6	6	Okney	w6554	8	53199	2.4	24

We specify a random effect Poisson model

$$\begin{aligned}y_i &\sim \text{Poisson}(E_i \rho_i) \\ \log(\rho_i) &= b_0 + \beta_1 x_i + v_i \\ v_i &\sim \text{Normal}(0, \sigma_v^2)\end{aligned}$$

where x_i is the exposure to sun in each area. Note that b_0 , β_1 , $1/\sigma_v^2$ are given independent "noninformative" priors.

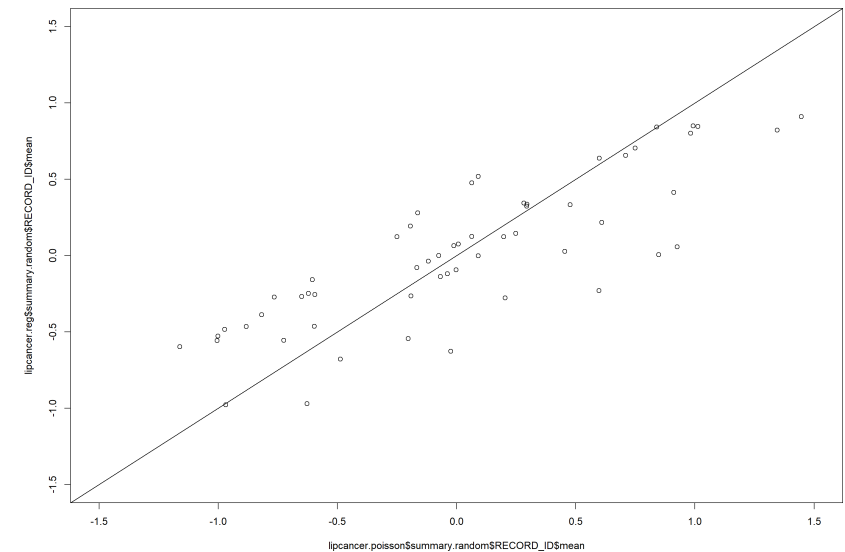
R-INLA code

```
> formula.reg.inla <- y ~ 1 + x +  
+   f(RECORD_ID,model="iid", hyper=list(prec=list(  
+     param=c(1,0.01))))  
>  
> lipcancer.reg <- inla(formula.reg.inla,family=''  
+                       data=LipCancer, E=E,  
+                       control.predictor=list(  
+                       control.compute=list(  
+                       control.fixed=list(me
```

```
              mean      sd 0.025quant      0.975quant  
(Intercept) -0.49119663 0.15719392 -0.8050305 -0.1773627  
x             0.06841513 0.01402741  0.0408243  0.0959999
```

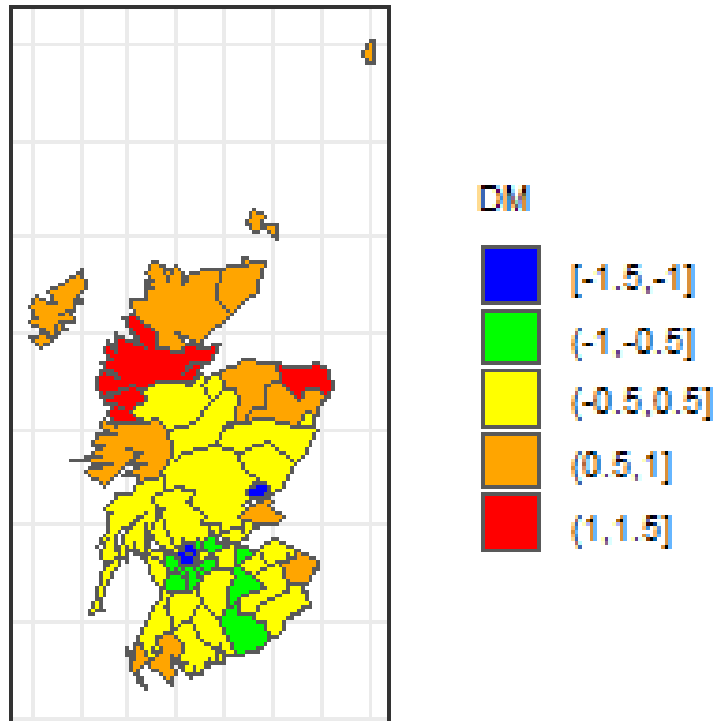
Plot the random effects for the disease mapping model and for the ecological regression

```
> plot(lipcancer.poisson$summary.random$RECORD_ID$mean  
> abline(0,1)
```

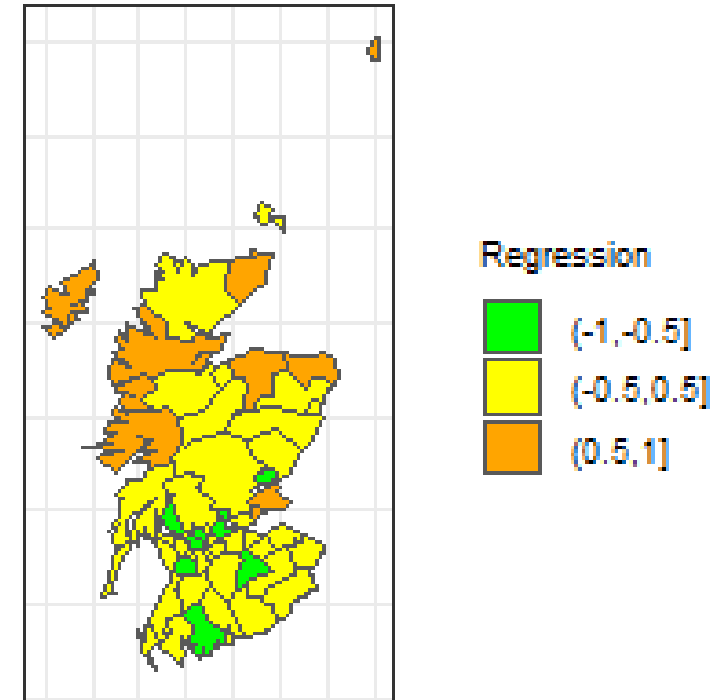


Comparing maps

Random effect v_i from disease mapping



Random effect v_i from ecological regression



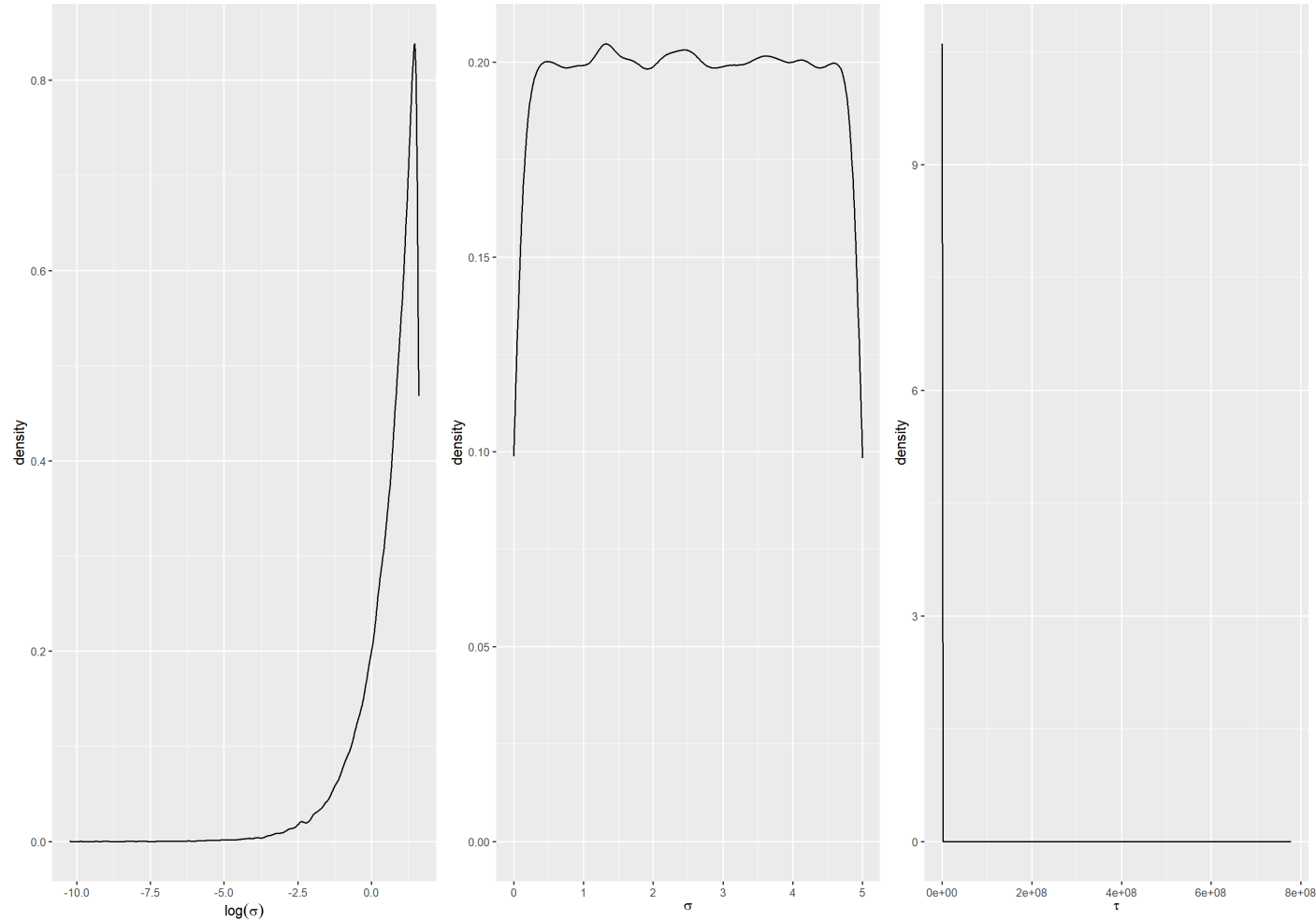
Choice of prior

How to specify priors?

- *Relatively* easy to specify priors on regression parameters
 - Typical choice is a Normal distribution
 - Tuning the variance it can be more or less informative
 - The scale of the variable it represents need to be considered
- Variances are more complex (and a bit more controversial)
- In small area studies we usually work with Poisson/Binomial distribution on data - no variance parameter; the main interest is on random effect variance.
- A Gamma (ϵ, ϵ) can be used on the precision but inference could be sensitive to choice of ϵ . Typically to ensure vague priors small ϵ are specified (e.g. 0.1, 0.01). However, this prior has also been criticised (e.g. (Gelman, 2006)) as it has a spike for values around 0.
- Careful as "non informative" prior distributions are sensitive to changes of scale.

Changing the scale

- For instance starting with a Uniform on the standard deviation we end up with a high density on low values for the precision



Remember...

- INLA parametrises the precision and the default is

$$\log(1/\sigma^2) \sim \text{logGamma}(1, 0.00005)$$

- However alternatives can be built, for instance:
 - Truncated Normal on log precision (`logtnormal`)
 - Uniform prior on the standard deviation: as it is not implemented we need to specify it through the expression as follows

```
UN.prior = "expression: log_dens = 0 - log(2) - theta / 2; return(log_dens);"
```

In general we need to be careful to check the level of information (weakly, strong) on the scale we are interested in (e.g. variance) and see what this corresponds on the standard deviation/precision (on which prior is usually specified).

See Gómez-Rubio (2020) for more information on how to specify priors in INLA.

Model selection

Which model?

All models are wrong, some models are useful.

G. Box

So the question is: how is my model doing?

- 1 in terms of model assumption
- 2 compared to other models

We can answer the first question using the **posterior predictive distribution**

We can answer the first question using **methods based on the trade-off between a measure of model fit and of model complexity**

Posterior predictive distribution

Main idea: If the combined model assumptions are reasonable, then our posterior model should be able to simulate data that's similar to the original one

- Let's assume we want to find the relationship between air pollution and asthma attacks and we collect data of the outcome over 500 days in a London Local Authority

We propose the following model (y =number of asthma attacks, x =level of PM_{10} in the previous 3 days $i = 1, \dots, 500$):

$$\begin{aligned}y_i &\sim \text{Poisson}(Pop\rho_i) \\ \log(\rho_i) &= b_0 + \beta x_i + v_i \\ b_0, \beta &\sim N(0, 0.001) \\ v_i &\sim N(0, \sigma_v^2) \\ \log(1/\sigma_v^2) &\sim \text{logGamma}(1, 0.00005)\end{aligned}$$

The assumptions are

- 1 that the data are distributed as **Poisson**
- 2 that there is a linear relationship between air pollution and the log number of asthma attacks
- 3 that the days are similar (we include a random effect)

Posterior predictive distribution

- We run the model and predict observations $y_1^*, \dots, y_{500}^*$ based on the posterior distribution of the parameters (note that we need to include `control.predictor` in the `inla` function to access these):

```
> asthma_formula1 <- y ~ x + as.factor(dow) + f(ID, model="iid")
> asthma_model1 <- inla(asthma_formula1, data=data, family="poisson",
+                       control.predictor=list(compute=TRUE), control.compute=list(config=TRUE,
+                                           waic=TRUE, return.marginals.predictor=TRUE))
```

Then to get the y_i values we need to incorporate an additional step getting a posterior sample from the fitted values using

```
> post <- inla.posterior.sample(100, asthma_model1)
> names(post[[1]])
```

```
[1] "hyperpar" "latent"  "logdens"
```

```
> post[[1]]$latent[1:3,]
```

```
Predictor:1 Predictor:2 Predictor:3
1.1431768  0.8692804  1.8645721
```

Note that:

- `post` is a list of 100 elements and each element includes a value from

1. the joint posterior distribution for the hyperparameters `post$hyperpar`
2. joint posterior distribution for the linear predictor η in `post$latent` (row 1 to N)
3. joint posterior distribution for the random effects v in `post$latent` (N + 1 to 2N)

Posterior predictive distribution

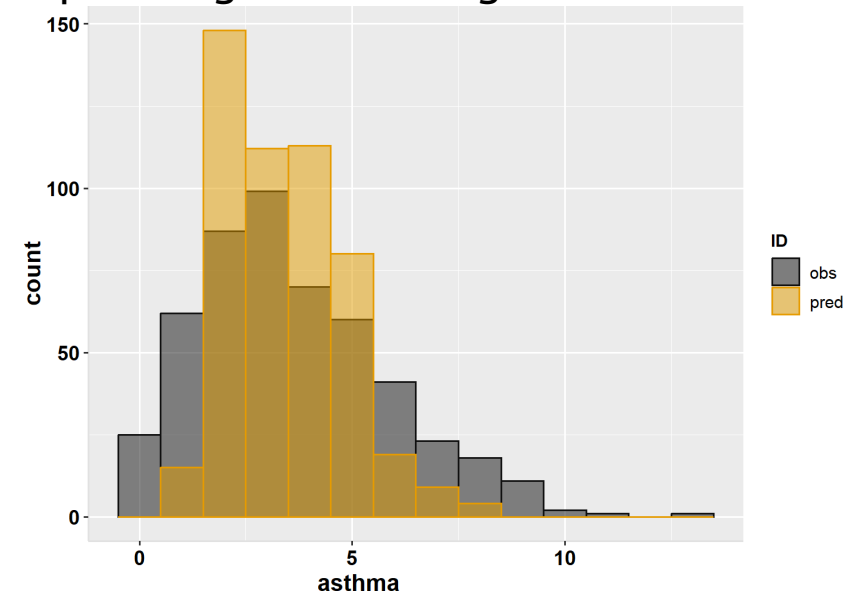
Then we sample from a Poisson where the mean is the output of the joint posterior sample (exponentiated to go back to the original scale), for the linear predictor

```
> #Exponentiate
> predlist <- do.call(cbind, lapply(post,
+                               function(X) exp(X$latent[startsWith(rownames(X$latent), "Pred")]))))
> #Sample from the Poisson
> pois.samples <- apply(predlist, 2, function(Z) rpois(n = length(Z), lambda = Z))
> pois.samples <- as.data.frame(pois.samples)
```

```
[1] 500 100
```

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10
1	3	4	2	8	2	0	7	1	3	2
2	4	0	3	2	4	5	2	3	2	0
3	4	4	4	7	3	7	5	5	5	3

Now we plot it against the original data



Posterior predictive distribution

Now let's assume we run a different model

$$\begin{aligned}y_i &\sim \text{Normal}(\theta_i, \sigma_2) \\ \text{logit}(\theta_i) &= b_0 + \beta x_i \\ b_0, \beta &\sim N(0, 0.001)\end{aligned}$$

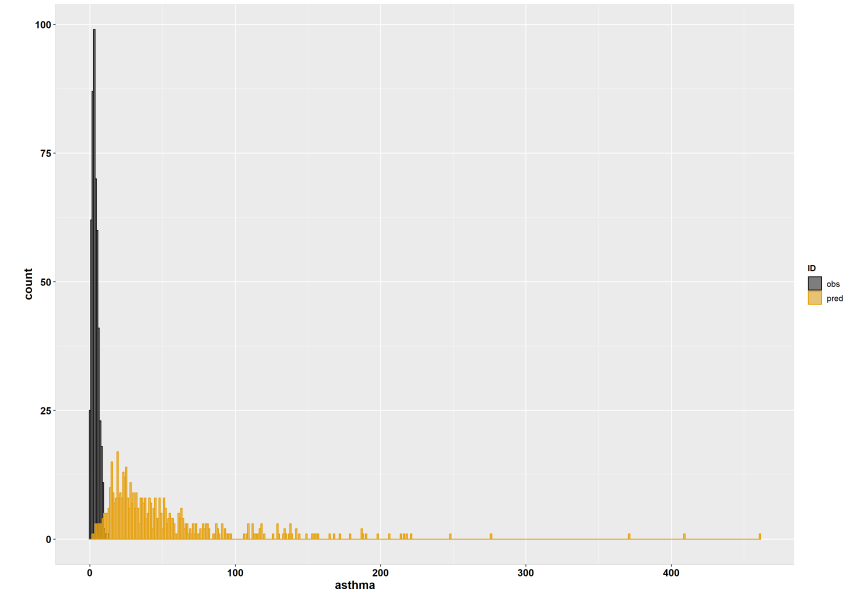
The assumptions are

- 1 that the data are distributed as **Gaussian**
- 2 that there is a linear relationship between air pollution and number of asthma attacks

Posterior predictive distribution

- We run the model and predict observations $y_1^*, \dots, y_{500}^*$ based on the posterior distribution of the parameters similarly to the previous model

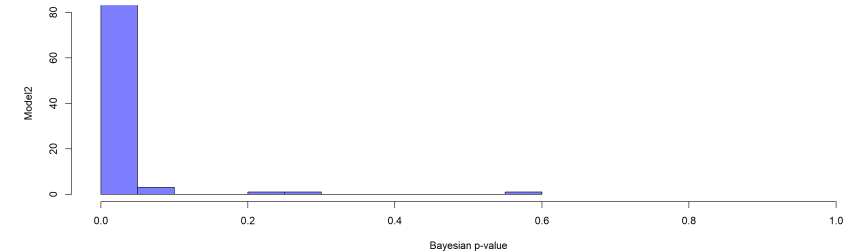
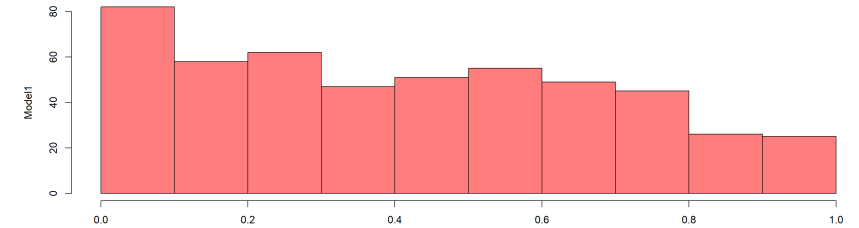
```
> asthma_formula2 <- y ~ x  
> asthma_model2 <- inla(asthma_formula2, data=dat  
+                       control.predictor=list(cc  
+                       control.compute=list(conf
```



Model comparison: Bayesian p-value

- We can use the posterior predictive distribution to compare to the observed one through a *p-value*

```
> # Model 1
> Bayesian_p <- matrix(NA, 500, 2)
>
> #Proportion of times the prediction is larger
> for(i in 1:500){
+   Bayesian_p[i,1] <- sum(pois.samples[i,]<y[i])/1000
+   Bayesian_p[i,2] <- sum(gaussian.samples[i,]<y[i])/1000
+ }
```



- Ideally we would expect a uniform distribution of the p-values which would tell us there is no pattern of over(under) estimation in the prediction
- Here model 1 seems far better than model 2

Model comparison: fit vs complexity

- When the interest lays mainly on the prior distribution or on the functional form of some parameters the deviance of the model can be used to evaluate the goodness of fit.

Given the data $\mathbf{y}$ with distribution $p(\mathbf{y} \mid \theta)$, the deviance of the model is defined as:

$$D(\theta) = -2\log p(\mathbf{y} \mid \theta)$$

where θ identifies the parameter of the likelihood

- Ex. $y_i \sim \text{Bernoulli}(\theta) \rightsquigarrow p(\mathbf{y} \mid \theta) = \prod_{i=1}^n \binom{n_i}{y_i} \theta^{y_i} (1 - \theta)^{n_i - y_i}$

$$D(\theta) = -2 \left[\sum_i y_i \log \theta_i + (n_i - y_i) \log(1 - \theta_i) + \log \binom{n_i}{y_i} \right]$$

Mean deviance

- The deviance of the model measures the variability linked to the likelihood, ie the probabilistic structure used for the observation (conditional on the parameters)
- This quantity is a random variable in the Bayesian framework, so it is possible to synthesise it through several indexes (mean, median, etc.)
- Many authors suggested using posterior mean deviance $(\overline{D}) = E_{\theta|y}[D(\theta)]$ as a measure of fit

DRAWBACK: more complex models will fit the data better and so will have smaller $\overline{D}$

- Need to have some measure of *model complexity* to trade off against $\overline{D}$

Watanabe AIC - WAIC

- Natural way to compare models is to use criterion based on trade-off between the fit of the data to the model and the corresponding complexity of the model
- Let m_i and v_i be the posterior predictive mean and variance for the i^{th} unit
- The effective model size is

$$p_W = \sum_{i=1}^n v_i$$

- The criteria is

$$WAIC = -2 \sum_{i=1}^n m_i + 2p_W$$

- The WAIC is readily available in INLA using `control.compute=list(waic=TRUE)`
- Linked to cross-validation
- WAIC has a model-fit and model-complexity components
- Smaller WAIC indicates the preferred model

Back to our example...

We run the model adding the waic (here for model 1, it is the same for model 2):

```
> asthma_model1 <- inla(asthma_formula1, data=data, family="poisson", control.predictor=list(link=1, compute=TRUE),  
+                        control.compute=list(waic=TRUE))
```

And now check the value of the WAIC

```
> # Poisson data distribution  
> asthma_model1$waic$waic
```

```
[1] 1956.986
```

```
> # Normal data distribution  
> asthma_model2$waic$waic
```

```
[1] 2163.642
```

The first model is still preferred as the WAIC is smaller.

Summary

- Hierarchical models allow **borrowing of strength** across units
 - posterior distribution of the unit-parameter borrows strength from the likelihood contributions for all the units, via their joint influence on the posterior estimates of the unknown hyper-parameters
 - improved efficiency
- Judgements of exchangeability need careful assessment → units suspected a priori to be systematically different might be modelled by including relevant covariates so that residual variability more plausibly reflects exchangeability
- Subgroups of prior interest should be considered separately

Careful on the prior specification

- non informative on one scale might be informative on another
- always run some sensitivity analyses changing the prior and investigating how this affect the estimates of parameters of interest
- posterior predictive distribution is useful to check if a model is in line with the data under study
- WAIC is a useful tool for model selection, easy to calculate in INLA

→ bear in mind that they can only be used to compare models - similarly to the AIC they do not have an absolute meaning.

References

- Blangiardo, M. and M. Cameletti (2015). *Spatial and spatio-temporal Bayesian models with R-INLA*. John Wiley & Sons.
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- Gómez-Rubio, V. (2020). *Bayesian inference with INLA*. CRC Press.